

Semantic Analysis

Compilers

Consider the statements:

$$\begin{aligned} S_1 &= x := 42; [y := 1; z := x + y] \\ S_2 &= x := \text{true}; [y := 1; z := x + y] \end{aligned}$$

Assume the following:

- x hashes to 0
- y hashes to 1
- z hashes to 2
- $\$$ is the initial stack
- $\oplus$ is the special “new block” marker

Analyze S_1 and S_2 using a symbol table and an auxiliary stack for scoping.

Solution

Here is the analysis of $x := 42; [y := 1; z := x + y]$:

AST	stack	symbol table	actions						
$x := 42; [y := 1; z := x + y]$	\$	<table><tr><td>0</td><td>1</td><td>2</td></tr></table>	0	1	2				
0	1	2							
$x := 42; [y := 1; z := x + y]$	\$	<table><tr><td>0</td><td>1</td><td>2</td></tr></table>	0	1	2				
0	1	2							
$x := 42; [y := 1; z := x + y]$	$\$x \oplus$	<table><tr><td>(x, int)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, int)			0	1	2	push x ; insert x ; push $\oplus$
(x, int)									
0	1	2							
$x := 42; [y := 1; z := x + y]$	$\$x \oplus$	<table><tr><td>(x, int)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, int)			0	1	2	
(x, int)									
0	1	2							
$x := 42; [y := 1; z := x + y]$	$\$x \oplus$	<table><tr><td>(x, int)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, int)			0	1	2	
(x, int)									
0	1	2							
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(x, int)									
0	1	2							
$x := 42; [y := \underline{1}; z := x + y]$	$\$x \oplus y$	<table><tr><td>(x, int)</td><td>(y, int)</td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, int)	(y, int)		0	1	2	push y ; insert y
(x, int)	(y, int)								
0	1	2							
$x := 42; [y := 1; z := x + y]$	$\$x \oplus y$	<table><tr><td>(x, int)</td><td>(y, int)</td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, int)	(y, int)		0	1	2	
(x, int)	(y, int)								
0	1	2							
$x := 42; [y := 1; z := \underline{x} + y]$	$\$x \oplus y$	<table><tr><td>(x, int)</td><td>(y, int)</td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, int)	(y, int)		0	1	2	
(x, int)	(y, int)								
0	1	2							
$x := 42; [y := 1; z := \underline{x} + y]$	$\$x \oplus y$	<table><tr><td>(x, int)</td><td>(y, int)</td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, int)	(y, int)		0	1	2	lookup x
(x, int)	(y, int)								
0	1	2							
$x := 42; [y := 1; z := x + y]$	$\$x$	<table><tr><td>(x, int)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, int)			0	1	2	lookup y ; push z ; insert z ; pop z, y
(x, int)									
0	1	2							

Here is the analysis of $x := \text{true}; [y := 1; z := x + y]$:

AST	stack	symbol table	actions						
$x := \text{true}; [y := 1; z := x + y]$	\$	<table><tr><td>0</td><td>1</td><td>2</td></tr></table>	0	1	2				
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	\$	<table><tr><td>0</td><td>1</td><td>2</td></tr></table>	0	1	2				
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x \oplus$	<table><tr><td>(x, bool)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)			0	1	2	push x ; insert x ; push $\oplus$
(x, bool)									
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x \oplus$	<table><tr><td>(x, bool)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)			0	1	2	
(x, bool)									
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x \oplus$	<table><tr><td>(x, bool)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)			0	1	2	
(x, bool)									
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x \oplus$	<table><tr><td>(x, bool)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)			0	1	2	
(x, bool)									
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x \oplus y$	<table><tr><td>(x, bool)</td><td>(y, int)</td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)	(y, int)		0	1	2	push y ; insert y
(x, bool)	(y, int)								
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x \oplus y$	<table><tr><td>(x, bool)</td><td>(y, int)</td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)	(y, int)		0	1	2	
(x, bool)	(y, int)								
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x \oplus y$	<table><tr><td>(x, bool)</td><td>(y, int)</td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)	(y, int)		0	1	2	
(x, bool)	(y, int)								
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x \oplus y$	<table><tr><td>(x, bool)</td><td>(y, int)</td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)	(y, int)		0	1	2	lookup x
(x, bool)	(y, int)								
0	1	2							
$x := \text{true}; [y := 1; z := x + y]$	$\$x$	<table><tr><td>(x, bool)</td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>2</td></tr></table>	(x, bool)			0	1	2	lookup y ; fail because $+ \not\in \mathbb{B} \rightarrow \mathbb{Z} \rightarrow T$
(x, bool)									
0	1	2							